

# Einstein's General Theory of Relativity

## Lecture 2: Gauss's Law, Accelerated Frames, and Curvature

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## Chapter 1

# Gauss's Law, Accelerated Frames, and Curvature

### 1.1 A Detour Through Dark Energy

Before returning to Newtonian gravity, let us settle a question left over from the preceding discussion. A positive cosmological constant contributes a very weak repulsive effect whose magnitude grows with separation. For the limited purpose at hand, a spring supplies a useful model. If its ordinary restoring force is

$$F_{\text{spring}} = -kr, \quad (1.1)$$

then the cosmological term may be represented schematically by

$$F_{\text{total}} = -(k - \lambda)r = -kr + \lambda r, \quad 0 < \lambda \ll k. \quad (1.2)$$

The analogy is not meant to turn an atom into a literal spring. It isolates the relevant point: adding a minute repulsive coefficient weakens a bound system slightly. Its equilibrium dimensions shift by a negligible amount; it is not torn apart. The same conclusion applies to laboratory systems, the solar system, and ordinary galactic binding. The effect becomes competitive only over cosmological distances.

This also separates an ordinary cosmological constant from the proposed *Big Rip*. A genuine constant does not grow in time. The rip scenario instead assumes an effective repulsion that becomes progressively stronger, first overcoming galactic binding, then stellar and planetary binding, and eventually microscopic binding. That is a different hypothesis, and the lecture treats a growing  $\lambda$  as physically implausible. A decreasing dark-energy component is conceptually distinct and is not ruled out by the same argument.

**Question from the audience.** Is  $\lambda$  a function of the mass, and would it still exist if no particles were present? <sup>a</sup>

**Response.** Here  $\lambda$  is a numerical constant, not a function of a particular source mass. It specifies how particles would respond if they were present. If there are no particles, there are simply no test bodies on which to display that force.

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<sup>a</sup>Lecture timestamp: 00:08:47–00:09:16.

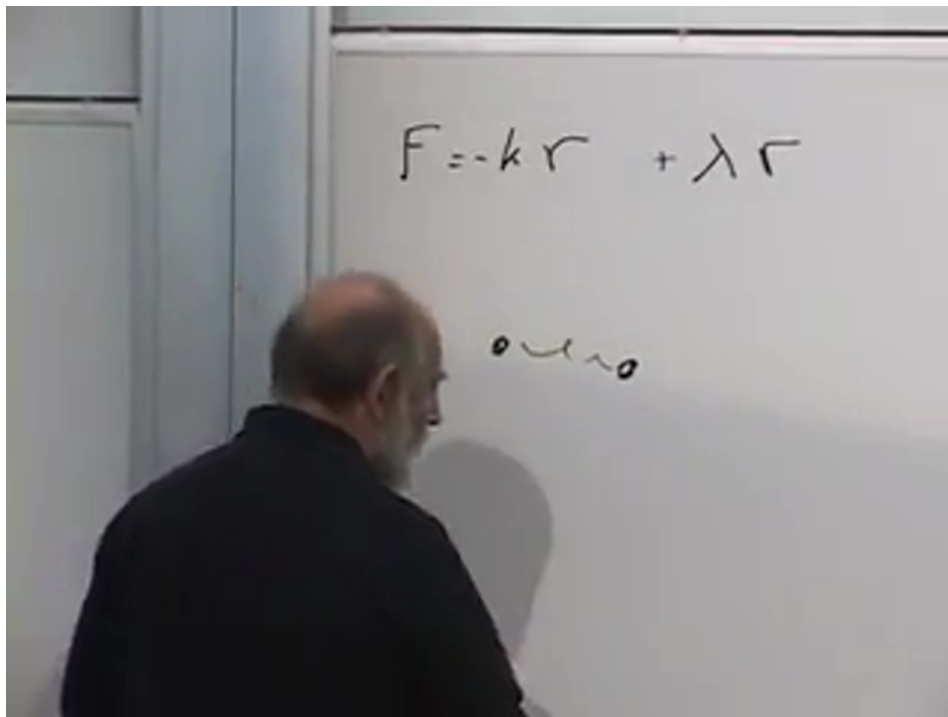


Figure 1.1: A spring supplies the opening model for a tiny repulsive cosmological contribution. The important comparison is between the strong restoring coefficient  $k$  and the exceedingly small coefficient  $\lambda$ .

*Lecture frame: 00:05:00.*

**Question from the audience.** Does the cosmological constant evolve with time? <sup>a</sup>

**Response.** In the standard interpretation it is constant, which is why the name is appropriate. More general forms of dark energy can vary. The opening argument distinguishes a possible decrease from the strongly disfavored increase required by the Big Rip.

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<sup>a</sup>Lecture timestamp: 00:09:16–00:09:49.

**Question from the audience.** How can dark energy be a large fraction of the cosmic energy budget if its local effect is so small? <sup>a</sup>

**Response.** Its density in a cubic meter is extraordinarily small. What matters cosmologically is that this small density fills an enormous volume smoothly. Accumulated over distances comparable with the observable universe, its gravitational effect can dominate the expansion even though it is irrelevant in the laboratory, the solar system, and most individual astronomical systems. For the same reason it is not a practical reservoir from which to extract a useful tankful of energy.

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<sup>a</sup>Lecture timestamp: 00:09:55–00:12:36.

**Question from the audience.** Is dark energy a relative energy, so that only differences in it matter? <sup>a</sup>

**Response.** The word *relative* needs a more precise definition before that question has a unique answer. The point retained here is that dark energy gravitates: it acts as a smooth source in the large-scale gravitational dynamics.

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<sup>a</sup>Lecture timestamp: 00:12:40–00:13:20.

**Question from the audience.** Is there a particle associated with dark energy? <sup>a</sup>

**Response.** Not in the cosmological-constant picture used here. Dark energy is not being described as a gas of particles. This is quite different from dark matter, which is generally believed to consist of particles.

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<sup>a</sup>Lecture timestamp: 00:13:25–00:14:01.

That answers the questions which motivated the detour. We now need a little more Newtonian gravity before we can move on to Einstein.

## 1.2 Del, Gradients, and Divergences

The necessary mathematics is mostly notation. Introduce the differential operator

$$\nabla = \left( \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right). \quad (1.3)$$

It resembles a vector because it has three components, but those components are operations rather than numbers.

When  $\phi(x, y, z)$  is a scalar field, applying del produces its gradient, a vector field:

$$\nabla\phi = \left( \frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z} \right). \quad (1.4)$$

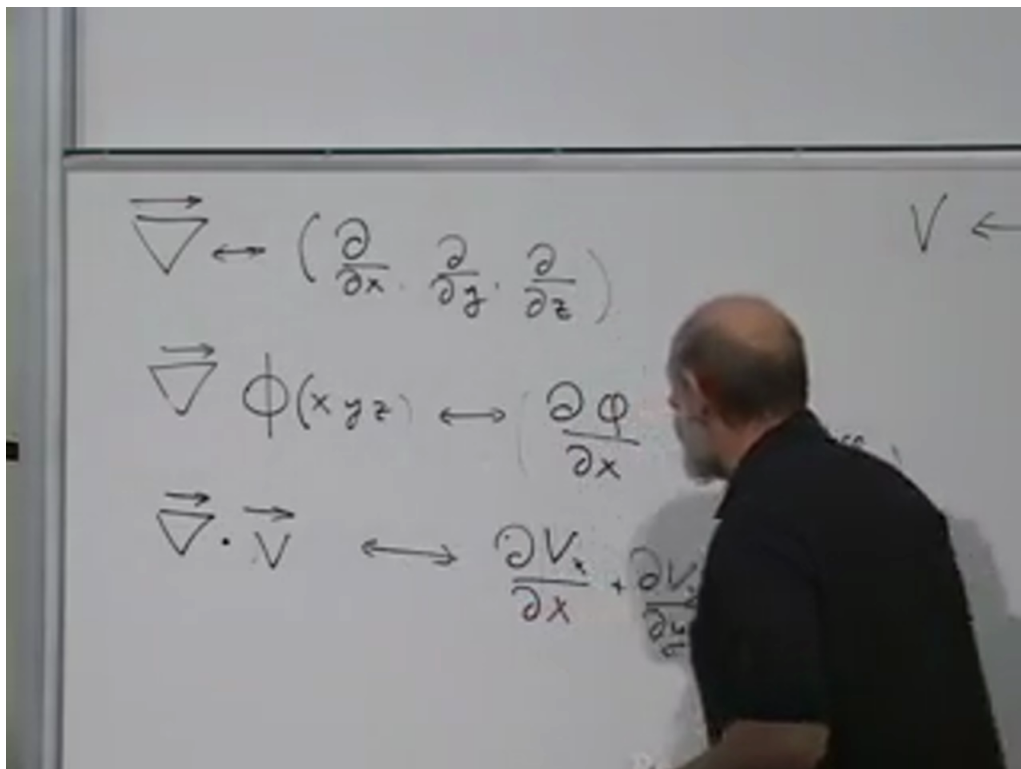


Figure 1.2: The del operator, the gradient of a scalar, and the divergence of a vector are assembled on one board. These operations will turn Newton's inverse-square law into a local field equation.

*Lecture frame: 00:19:30.*

For a vector field  $\vec{v} = (v_x, v_y, v_z)$ , taking the dot product with del gives the divergence,

$$\nabla \cdot \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}. \quad (1.5)$$

Just as the dot product of two ordinary vectors is a scalar, the divergence of a vector field is a scalar field. A fleeting verbal reversal at the board does not change this type: gradient maps scalar to vector; divergence maps vector to scalar.

A field is simply a quantity assigned at every point. If the assigned quantity is a number, it is a scalar field; if it has magnitude and direction, it is a vector field. The gravitational acceleration is our first important vector field.

### 1.3 The Gravitational Acceleration Field

Imagine source masses  $m_i$  at positions  $\vec{x}_i$ . To define the field at  $\vec{x}$ , place a test mass there, release it, and record its acceleration. The test mass is chosen small enough that its own disturbance of the sources can be neglected. Define

$$\vec{R}_i = \vec{x}_i - \vec{x}, \quad R_i = |\vec{R}_i|, \quad \hat{R}_i = \frac{\vec{R}_i}{R_i}. \quad (1.6)$$

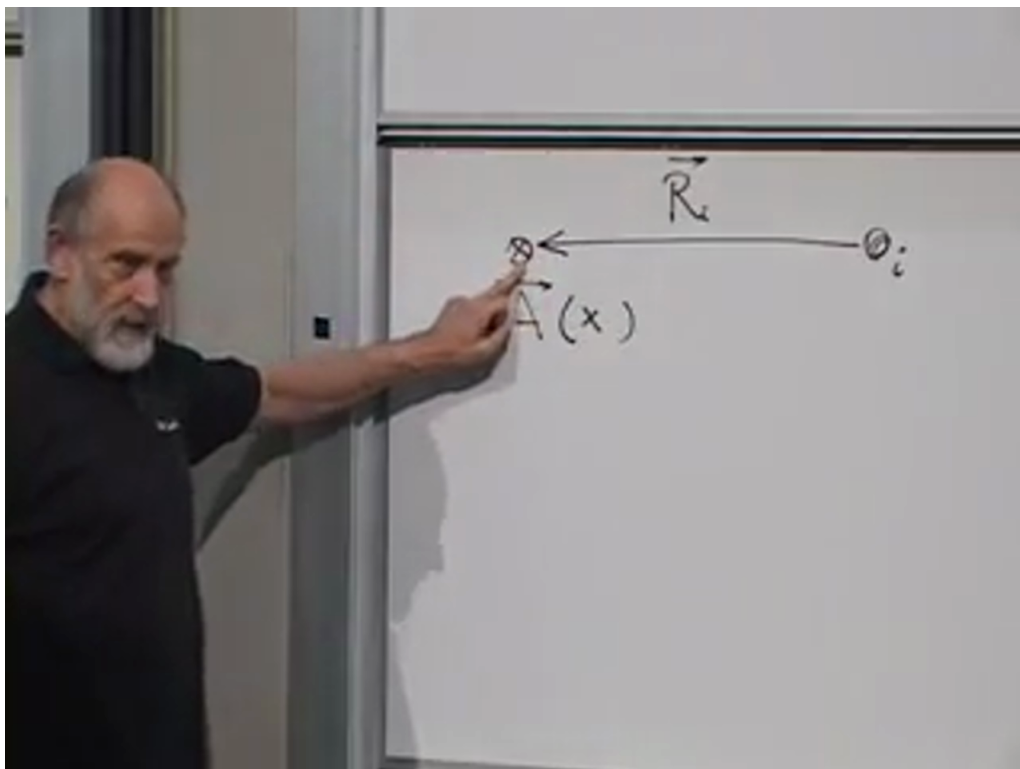


Figure 1.3: Source masses, a test point, and the displacement vector used to fix the direction of every term in the gravitational acceleration field.

Lecture frame: 00:23:26.

Thus  $\vec{R}_i$  points from the field point toward source  $i$ . This choice resolves the sign hesitation that occurs while the diagram is being drawn: attraction is then along  $+\vec{R}_i$ , and no extra minus sign is needed.

The acceleration field is

$$\vec{A}(\vec{x}) = G \sum_i m_i \frac{\vec{R}_i}{R_i^3} = G \sum_i m_i \frac{\hat{\vec{R}}_i}{R_i^2}. \quad (1.7)$$

The unit-vector form displays the inverse-square magnitude. The  $\vec{R}_i/R_i^3$  form packages magnitude and direction into a single vector expression. Moving the test point changes every  $\vec{R}_i$ , so  $\vec{A}$  depends on position and is indeed a field.

For a continuous distribution we replace a large collection of microscopic sources by the mass density

$$\rho(\vec{x}) = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V}. \quad (1.8)$$

The density is itself a scalar field. Newton's many-source equation can now be summarized by a much more compact local statement.

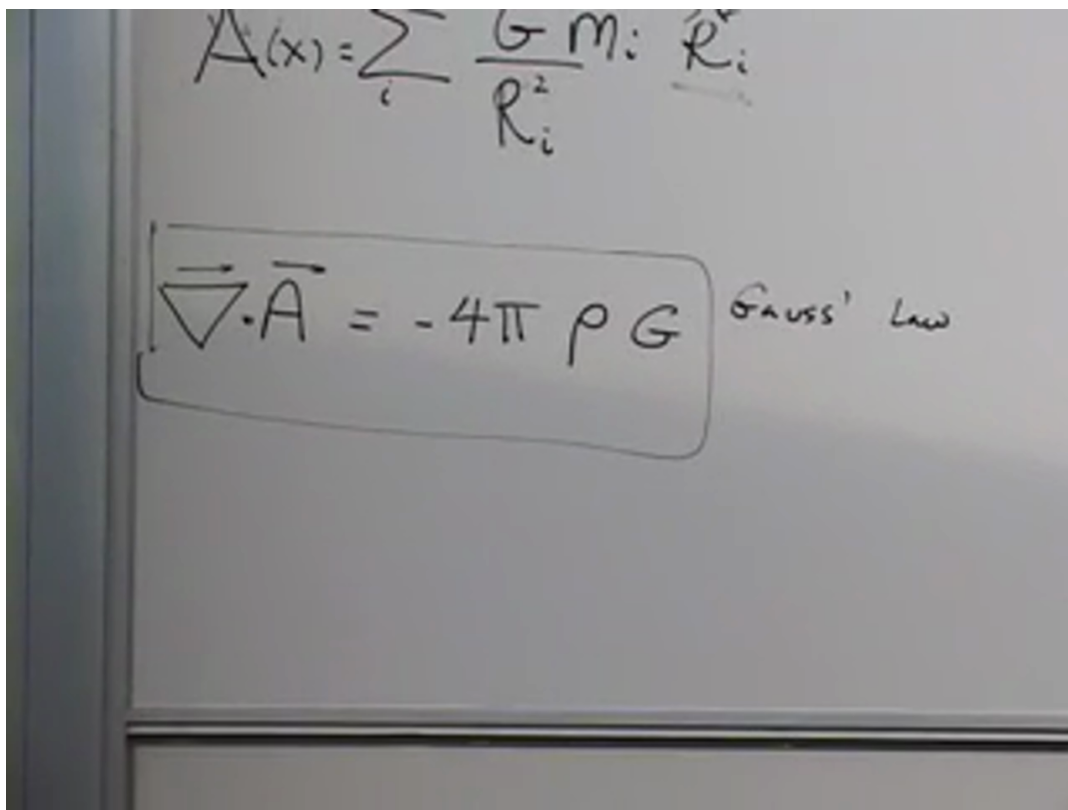


Figure 1.4: The many-source acceleration field and its compact local replacement,  $\nabla \cdot \vec{A} = -4\pi G\rho$ .

Lecture frame: 00:32:20.

## 1.4 Gauss's Law and Gauss's Theorem

The field equation for Newtonian gravity is

$$\boxed{\nabla \cdot \vec{A} = -4\pi G\rho.} \quad (1.9)$$

Positive mass makes the field point inward. What would be positive outward divergence for a source is therefore negative here: an inward convergence. Equation (1.9) is *Gauss's law for gravity*. It is a physical law, equivalent to Newton's relation between mass and gravitational acceleration.

Gauss's *theorem* is different. It is a mathematical identity valid for any sufficiently regular vector field and any region  $V$  bounded by the closed surface  $\partial V$ :

$$\boxed{\int_V \nabla \cdot \vec{A} dV = \oint_{\partial V} \vec{A} \cdot \hat{n} d\sigma.} \quad (1.10)$$

The region need not be spherical. Subdivide its volume into tiny cells, evaluate the divergence in each, multiply by the cell volume, and add. Internal fluxes cancel between neighboring cells, leaving only the outward normal component  $A_\perp = \vec{A} \cdot \hat{n}$  on the boundary.

The fluid analogy makes the theorem immediate. If  $\vec{A}$  were a velocity field, the integrated divergence would measure the net amount of fluid created inside the region, and the surface integral would measure the amount flowing out. Gauss's theorem equates these two descriptions. In gravity we combine this mathematical theorem with the physical source law (1.9).

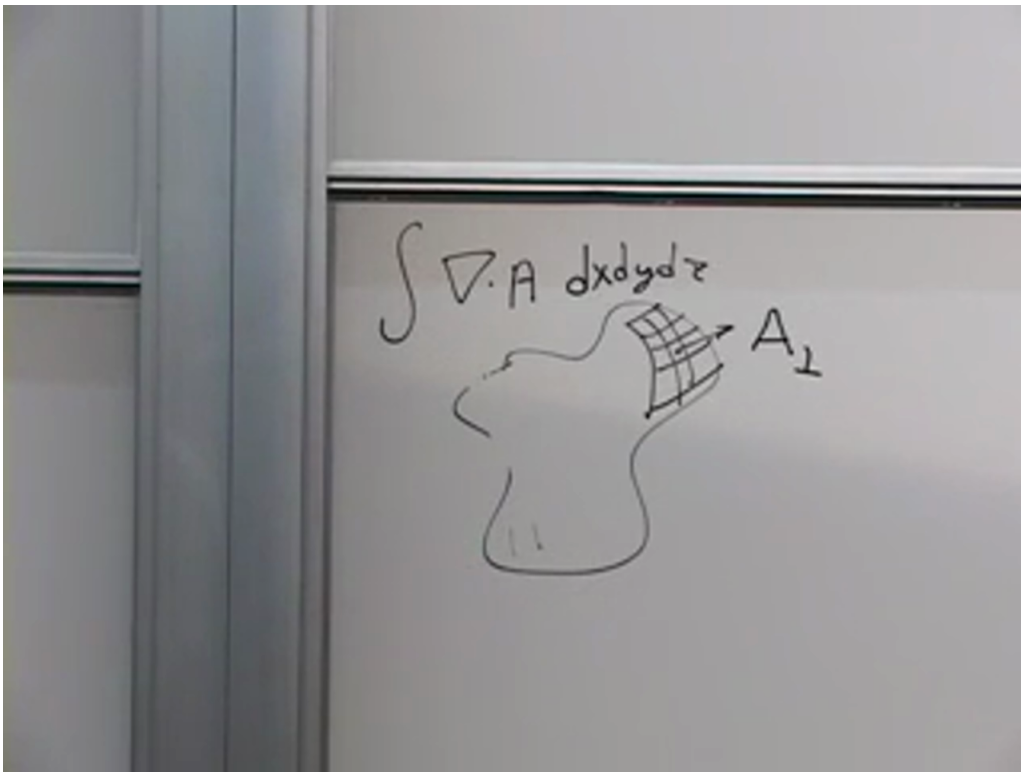


Figure 1.5: Gauss's theorem converts the integral of a volume divergence into the outward flux through the bounding surface. The marked surface patch carries area  $d\sigma$  and normal component  $A_{\perp}$ .

*Lecture frame: 00:35:00.*

## 1.5 Spherical Symmetry and the Exterior Field

Take a spherically symmetric, or isotropic, mass distribution and surround it with a concentric Gaussian sphere of radius  $r$ . Integrating Gauss's law gives

$$\int_V \nabla \cdot \vec{A} dV = -4\pi G \int_V \rho dV = -4\pi G M_{\text{enc}}(r). \quad (1.11)$$

Spherical symmetry forces the field to be radial and to have the same radial component everywhere on the Gaussian sphere. If  $\hat{r}$  points outward,

$$\oint_{\partial V} \vec{A} \cdot \hat{n} d\sigma = A_r(r) 4\pi r^2. \quad (1.12)$$

Equating the two sides,

$$4\pi r^2 A_r = -4\pi G M_{\text{enc}}(r), \quad (1.13)$$

$$A_r(r) = -\frac{G M_{\text{enc}}(r)}{r^2}.$$

When the Gaussian sphere lies outside the whole body,  $M_{\text{enc}} = M$ . Its exterior field is therefore exactly the field of a point mass  $M$  at the center. This derivation also explains the convenient  $4\pi$  in Gauss's law: it cancels the area  $4\pi r^2$  of a sphere.

The compact divergence equation is equivalent to the cumbersome sum over all source particles, but in a symmetric problem it does the calculation almost at once.

## 1.6 Inside a Uniform Earth

The same method answers the question of the field inside the Earth. Choose a Gaussian sphere of radius  $r < R$  within a model Earth of radius  $R$  and uniform density  $\rho$ . The real Earth is denser near its center, but uniform density is sufficient to reveal the structure. The enclosed mass is

$$M_{\text{enc}}(r) = \frac{4\pi}{3} \rho r^3. \quad (1.14)$$

Substitution into Equation (1.13) gives

$$A_r(r) = -\frac{4\pi G \rho}{3} r, \quad r < R.$$

(1.15)

At  $r = R$ , this agrees with the exterior result because  $M = (4\pi/3)\rho R^3$ .

Multiplying the acceleration by a test mass  $m$  gives

$$m\ddot{r} = -m \frac{4\pi G \rho}{3} r. \quad (1.16)$$

This is a harmonic oscillator with

$$\omega^2 = \frac{4\pi G \rho}{3} = \frac{GM}{R^3} = \frac{g_s}{R}, \quad (1.17)$$

where  $g_s = GM/R^2$  is the surface acceleration. A frictionless tunnel through the idealized uniform Earth would therefore produce an oscillation through the center and back.

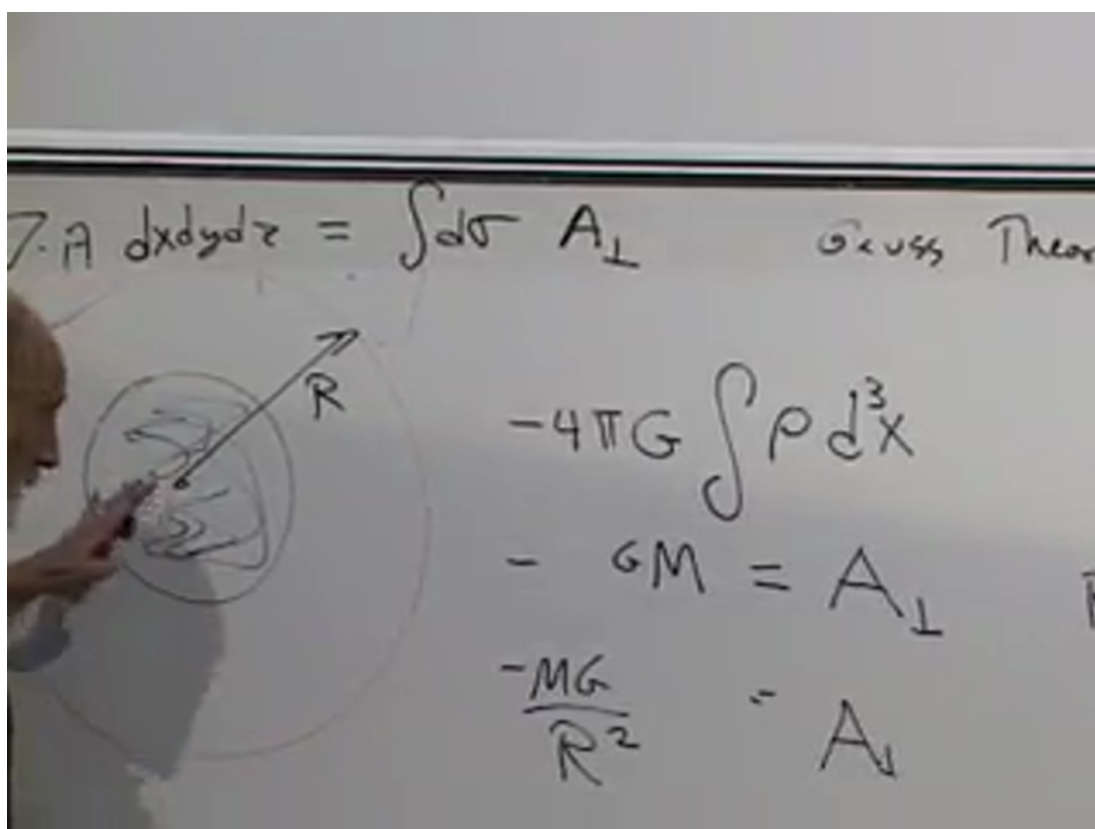


Figure 1.6: For an isotropic source, the volume integral gives the enclosed mass while the surface integral becomes  $4\pi r^2 A_r$ , recovering the exterior inverse-square field.

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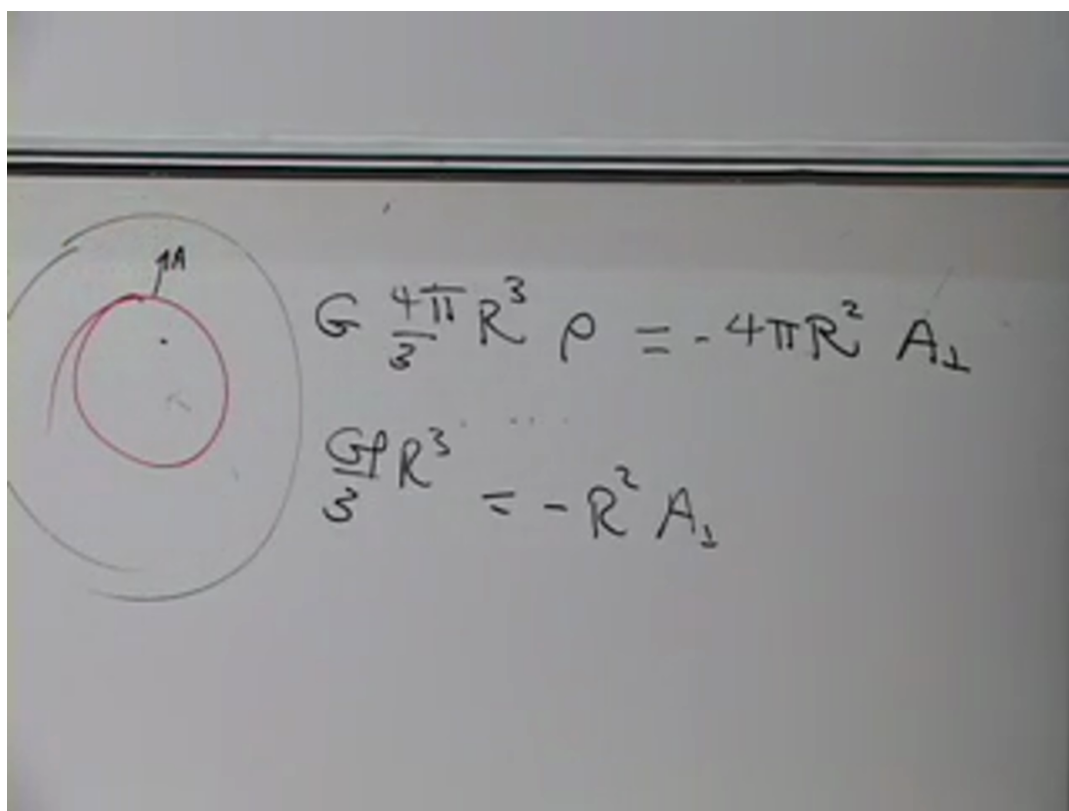


Figure 1.7: The inner Gaussian sphere and the rederived enclosed-mass calculation. Keeping the volume factor  $4\pi r^3/3$  and the  $4\pi$  in Gauss's law yields  $A_r = -(4\pi G\rho/3)r$ .

Lecture frame: 00:57:00.

The lecture makes an uncertain ten-minute estimate. Evaluating the model instead gives

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{R}{g_s}} \approx 84 \text{ min.} \quad (1.18)$$

The trip from one surface to the opposite surface is half a period, about 42 minutes; the fall from the surface to the center is about 21 minutes. The numerical correction leaves the point of the example unchanged: Gauss's law solves a many-body problem which would be forbidding as a direct sum.

**Question from the audience.** Why can the matter outside the Gaussian sphere be neglected?

<sup>a</sup>

**Response.** It is not merely negligible; for spherical shells its net field at every interior point is exactly zero. Contributions from the different parts of an outer shell cancel. Thus only enclosed mass enters  $A_r(r)$ . A thin spherical shell has zero field inside and the same exterior field as a point mass at its center. Newton established this geometrically before the flux formulation made the cancellation immediate.

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<sup>a</sup>Lecture timestamp: 00:48:20–00:50:20.

**Question from the audience.** Is the gravitational field continuous across a shell? <sup>a</sup>

**Response.** Across an ideal shell of zero thickness and finite surface mass density, the normal component of the field jumps. A shell with finite thickness replaces that jump by a smooth change through the material. The ideal discontinuity is the limiting form of Gauss's law.

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<sup>a</sup>Lecture timestamp: 00:53:02–00:53:40.

**Question from the audience.** If enclosed mass grows as  $r^3$ , why does the interior field grow only linearly with  $r$ ? <sup>a</sup>

**Response.** The flux also contains the Gaussian-sphere area, proportional to  $r^2$ . Hence

$$A_r r^2 \propto M_{\text{enc}} \propto r^3 \quad \implies \quad A_r \propto r.$$

The spoken conclusion  $1/r$  during the live algebra is a slip; the rederivation immediately returns to the correct linear field. The coefficient is fixed by retaining every  $4\pi$ :  $A_r = -(4\pi G\rho/3)r$ .

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<sup>a</sup>Lecture timestamp: 00:53:40–00:57:12.

Gravitational potential is announced at this point but deliberately postponed. We move instead from this old physics to the equivalence principle.

## 1.7 The Accelerating Elevator

Let an elevator accelerate horizontally with magnitude  $g$ . A released object can remain at rest in an inertial frame outside while the elevator moves beneath it. The passenger describes the same event differently: the object accelerates toward the floor with acceleration  $-g$ . Every freely released body behaves in the same way, just as it would in a uniform gravitational field.

The observation sounds elementary. Einstein's step was to use it as a principle applying not merely to dropped masses but to all local physics. Before doing that, let us express the change of frame mathematically.

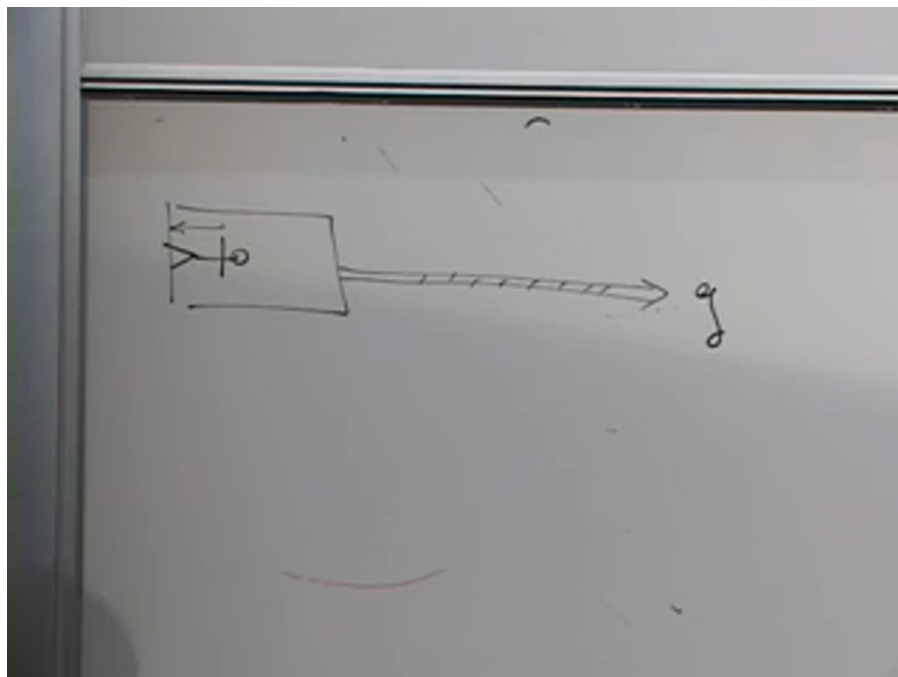


Figure 1.8: The horizontal elevator accelerates to the right. In the passenger's frame a released object accelerates toward the rear floor, producing the appearance of a uniform gravitational field.

*Lecture frame: 01:01:20.*

### 1.7.1 Uniform Velocity

Place  $x$  horizontally and  $t$  vertically on a spacetime diagram. Suppose first that the elevator moves with constant velocity  $v$ , and call its floor  $x' = 0$ . Newtonian coordinates are related by the Galilean transformation

$$x' = x - vt, \quad t' = t. \quad (1.19)$$

The equality  $t' = t$  belongs to Newtonian physics and will be modified in special relativity.

If a ball is at rest outside, then  $x$  is constant, and

$$\dot{x}' = -v, \quad \ddot{x}' = 0. \quad (1.20)$$

The moving passenger sees the ball pass backward, but sees no acceleration. Uniform relative motion creates no gravitational field.

### 1.7.2 Uniform Acceleration

Now let the elevator start from rest and accelerate uniformly. Its floor follows

$$x_{\text{floor}}(t) = \frac{1}{2}gt^2, \quad (1.21)$$

a parabola in the  $x$ - $t$  diagram. Coordinates carried by the elevator are therefore

$$x' = x - \frac{1}{2}gt^2. \quad (1.22)$$

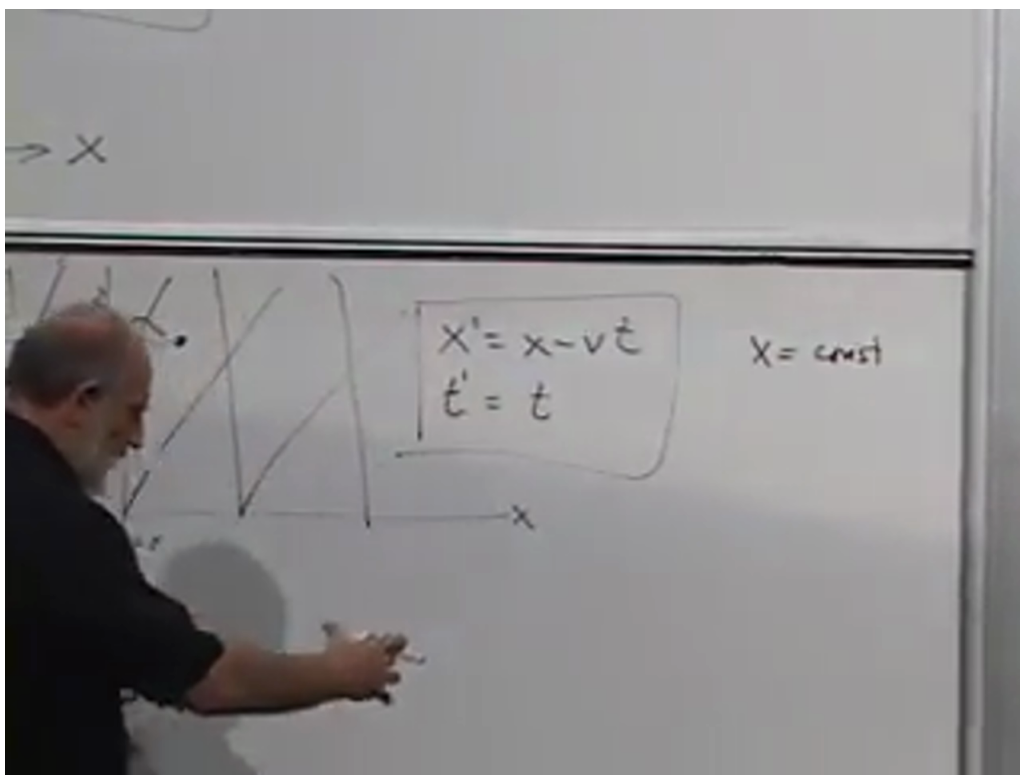


Figure 1.9: Straight worldlines for the stationary and uniformly moving coordinate grids, together with the Galilean transformation  $x' = x - vt$ ,  $t' = t$ .

*Lecture frame: 01:10:20.*

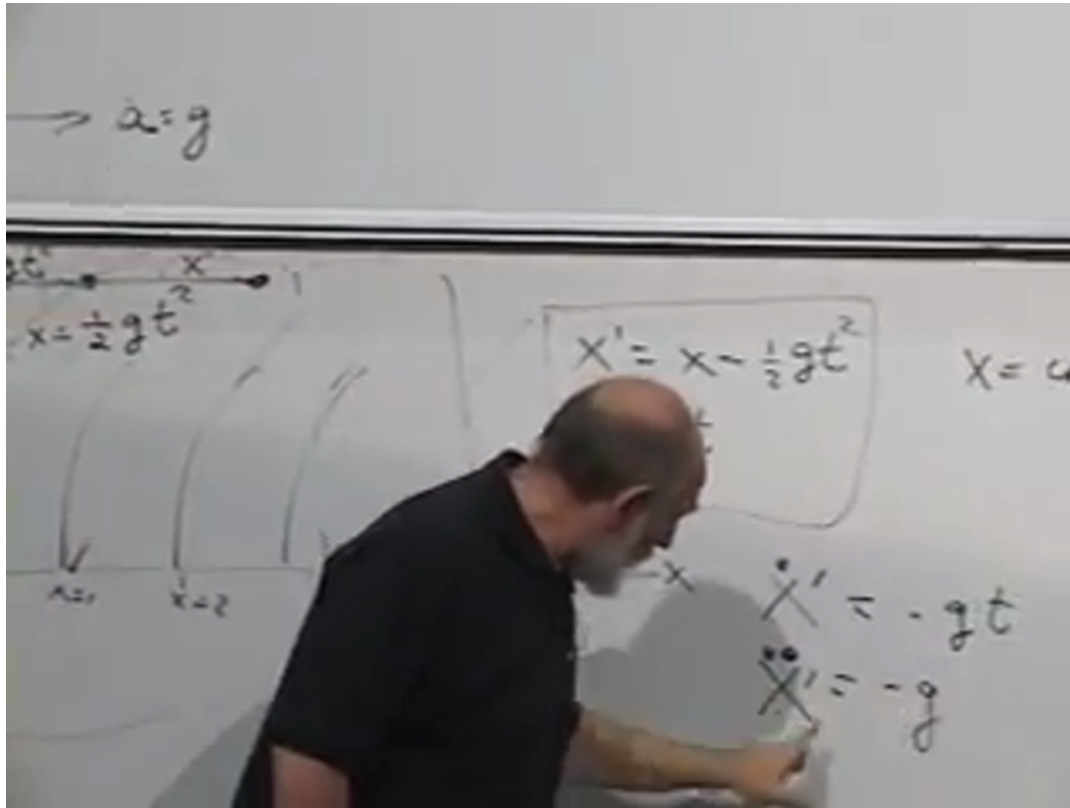


Figure 1.10: The accelerated coordinate transformation and its two time derivatives. Curved coordinate worldlines yield the effective acceleration  $\ddot{x}' = -g$ .

*Lecture frame: 01:16:20.*

For a ball at rest in the outside inertial frame,

$$\dot{x}' = -gt, \quad \boxed{\ddot{x}' = -g.} \quad (1.23)$$

This is the elevator argument in equations. An accelerated frame is a curvilinear coordinate system in spacetime, and in those coordinates a force-free body acquires the apparent acceleration of a uniform gravitational field.

This is the first bridge between gravity and curvature: acceleration is represented by curved coordinate lines. It is not yet spacetime curvature, because the underlying Newtonian spacetime has not changed. The distinction between curved coordinates and genuinely curved geometry will matter at the end of the lecture.

## 1.8 Light Falls

The equivalence principle is powerful only if it applies to every local law of physics. Consider an elevator accelerating upward. A horizontal beam is emitted at the instant the elevator begins to accelerate. In the outside inertial frame the light travels straight while the cabin rises. In the elevator frame the beam therefore strikes the far wall below its original height and traces a downward parabola. Its transverse acceleration is  $-g$ .

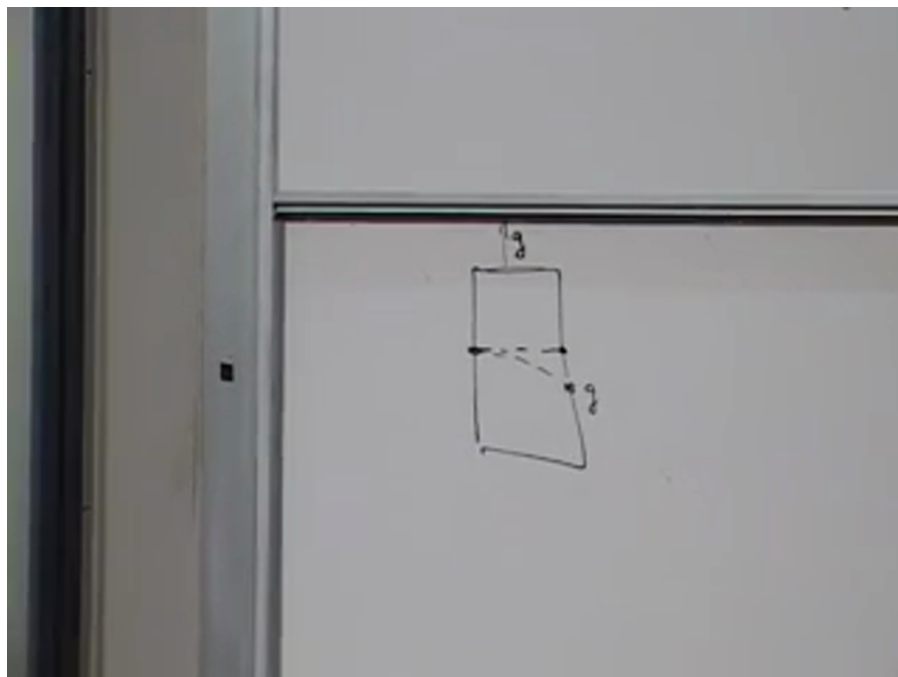


Figure 1.11: A horizontal light beam in an upward-accelerating elevator. The rising cabin makes the beam's path curve downward in the passenger's coordinates, showing that light falls in a gravitational field.

*Lecture frame: 01:20:20.*

**Question from the audience.** What happens if the light is sent along the direction of the elevator's acceleration? <sup>a</sup>

**Response.** The effect is then longitudinal rather than a transverse bending. It changes the light's energy, and therefore its frequency, rather than changing the locally measured speed  $c$ . This is the seed of the gravitational frequency-shift argument.

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<sup>a</sup>Lecture timestamp: 01:20:36–01:21:01.

### 1.8.1 A Crude Solar-Deflection Estimate

Take a ray that just skims the Sun, whose mass and radius are  $M$  and  $R$ . Near closest approach, approximate the downward acceleration by the surface value

$$g_{\odot} \approx \frac{GM}{R^2}. \quad (1.24)$$

The ray spends roughly one solar diameter in the strong-field region, so

$$\Delta t \approx \frac{2R}{c}. \quad (1.25)$$

Its acquired transverse velocity is then

$$\Delta v_{\perp} \approx g_{\odot} \Delta t \approx \frac{2GM}{Rc}. \quad (1.26)$$

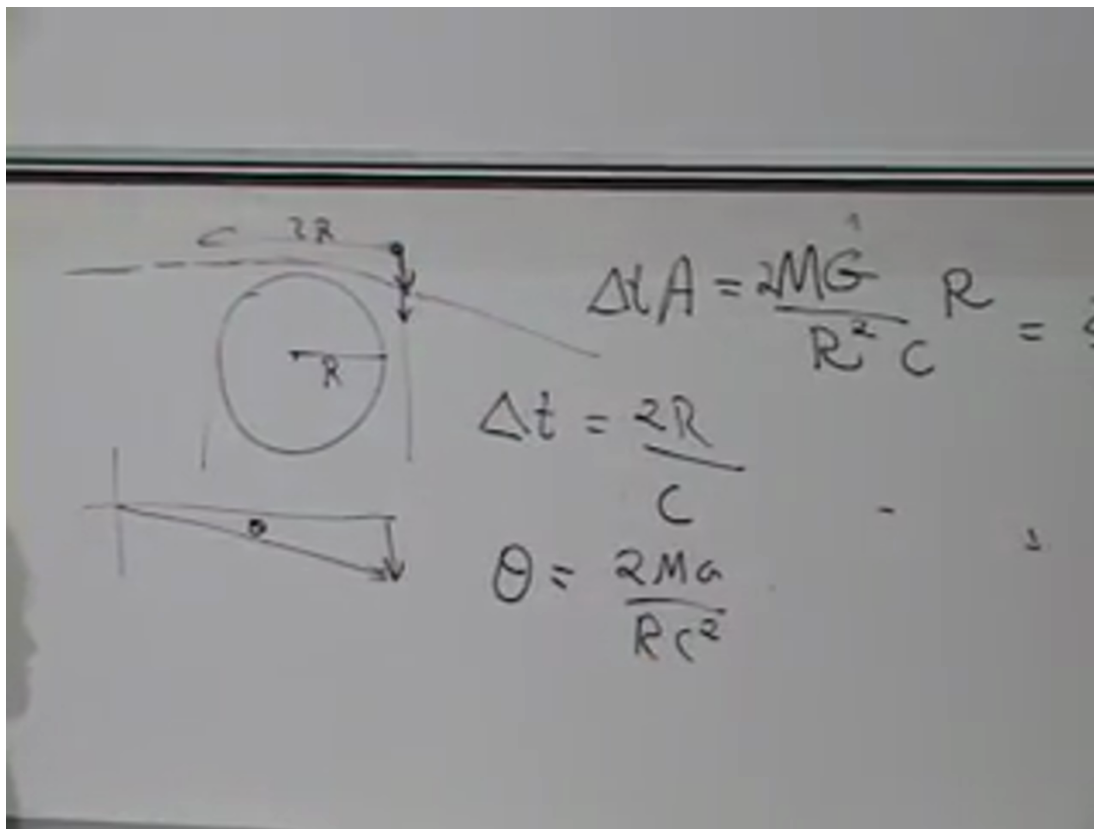


Figure 1.12: The sun-skimming ray estimate combines  $g_{\odot} \approx GM/R^2$  with a crossing time  $2R/c$  to obtain the weak-field deflection scale  $2GM/(Rc^2)$ .

Lecture frame: 01:27:20.

For a small angle,

$$\theta_{\text{Newton}} \approx \frac{\Delta v_{\perp}}{c} \approx \frac{2GM}{Rc^2}. \quad (1.27)$$

**Question from the audience.** Would ordinary Newtonian mechanics, applied to a photon with effective mass  $E/c^2$ , predict the observed bending? <sup>a</sup>

**Response.** It gives the estimate  $2GM/(Rc^2)$ , the same result obtained from this elementary equivalence argument. The full general-relativistic result is not half of this value but exactly twice it:

$$\theta_{\text{GR}} = \frac{4GM}{Rc^2}.$$

The extra factor reflects spatial curvature, which the Newtonian estimate does not include. The solar deflection was tested by comparing stellar positions near the Sun during an eclipse.

<sup>a</sup>Lecture timestamp: 01:21:03–01:28:22.

The historical point is larger than the rough numerical calculation. Gravity and acceleration had long been compared mechanically; Einstein applied the equivalence to light and electromagnetic phenomena, where its consequences were new.

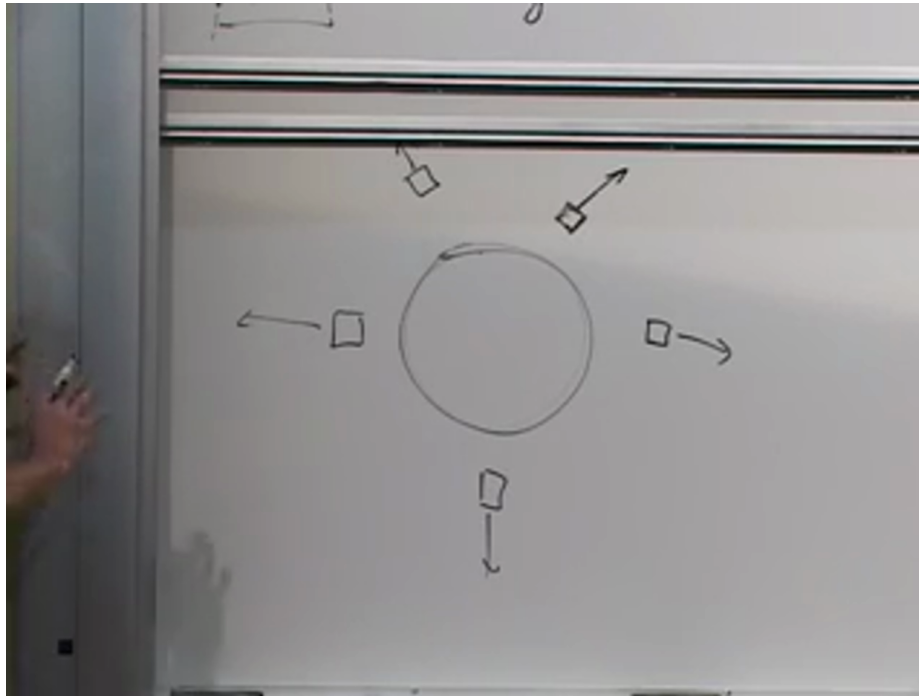


Figure 1.13: The Earth's radial field cannot be replaced globally by one uniformly accelerating elevator. Its changing direction and magnitude leave measurable relative accelerations.

*Lecture frame: 01:31:20.*

## 1.9 Why Equivalence Is Local

A single uniformly accelerated frame cannot reproduce the whole field of a round Earth. The gravitational acceleration points in different directions at different places and changes in magnitude with radius. It can be made uniform only within a region too small, and during a time too short, to resolve those variations.

A freely falling observer small enough to sample only one point can remove gravity locally. An extended observer cannot. The feet, being nearer the Earth, accelerate more strongly than the head; neighboring vertical trajectories spread or contract because their directions differ. The body is stretched radially and squeezed transversely. These relative accelerations are tidal forces.

Tidal forces are the invariant remainder of gravity in this argument. A uniform field has zero spatial variation and can be canceled by uniform acceleration. A realistic field has nonzero spatial derivatives; where mass is present, its divergence obeys  $\nabla \cdot \vec{A} = -4\pi G\rho$ , while tidal derivatives can remain nonzero even in the surrounding vacuum. No single accelerated coordinate system can remove those variations over an extended region.

**Question from the audience.** Does special relativity still hold inside the local elevator? <sup>a</sup>

**Response.** Yes. In a sufficiently small freely falling laboratory, local physics is special relativistic and inertial frames are related by Lorentz transformations. The coordinate calculations above used the Newtonian approximation only to establish the elementary elevator picture; the relativistic refinement is deferred.

<sup>a</sup>Lecture timestamp: 01:36:18–01:36:35.

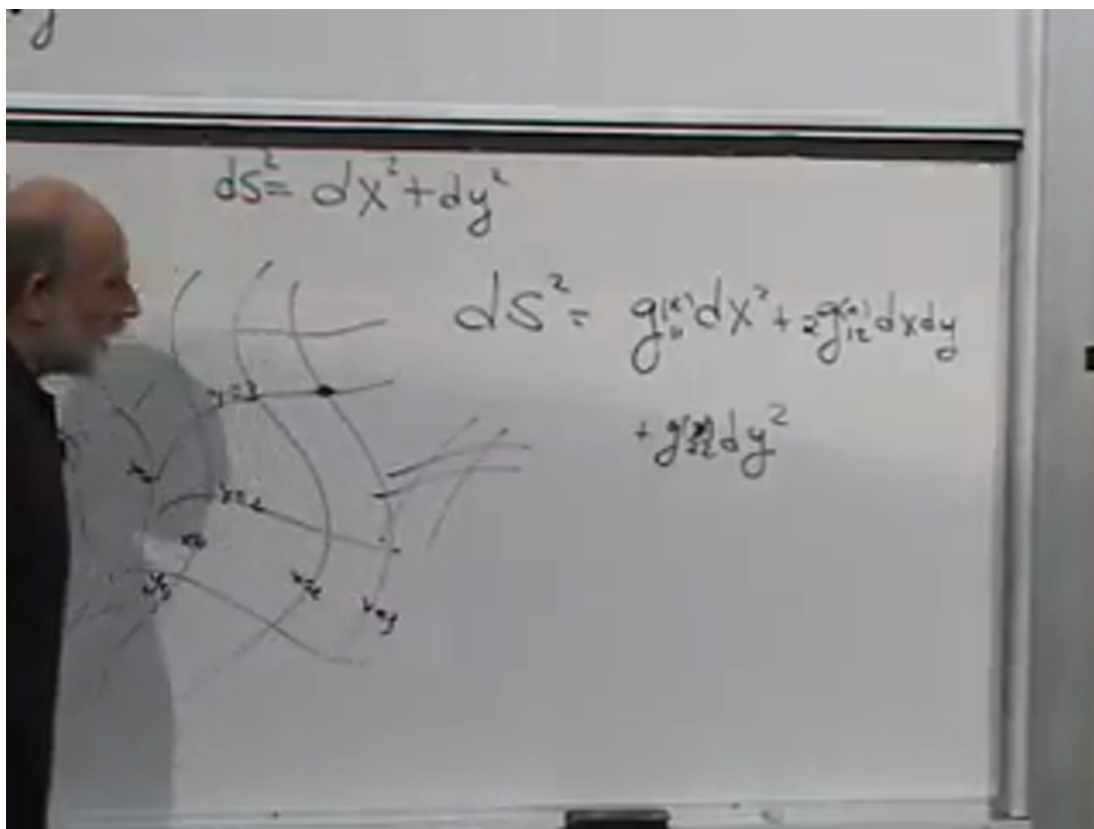


Figure 1.14: Cartesian and curvilinear coordinate meshes with their metric forms. A complicated metric can describe a flat board merely because the coordinates are complicated.

*Lecture frame: 01:42:20.*

## 1.10 The Metric and the Meaning of Curvature

We finish by isolating the geometric language needed later. A geometry is determined by the distances between neighboring points. In rectangular Cartesian coordinates on a flat blackboard, Pythagoras gives

$$ds^2 = dx^2 + dy^2. \quad (1.28)$$

The same flat board may be covered by curved coordinate lines. In general coordinates its infinitesimal distance takes the quadratic form

$$\boxed{ds^2 = g_{xx}(x, y) dx^2 + 2g_{xy}(x, y) dx dy + g_{yy}(x, y) dy^2.} \quad (1.29)$$

The three coefficient functions form the two-dimensional metric. The off-diagonal coefficient  $g_{xy}$  records the failure of the coordinate directions to be perpendicular. The diagonal coefficients record the local scale of coordinate intervals; if the grid stretches or compresses from place to place, they depend on position.

Curved coordinate lines do not by themselves imply curved geometry. A geometry is flat if some coordinate choice reduces its metric to the Euclidean form (1.28) throughout the region under consideration. If no coordinate transformation can do so, the geometry is intrinsically curved.

The examples make the distinction concrete:

- A plane is flat.
- A sphere is curved; no coordinate grid makes its metric globally Pythagorean.
- A cylinder is intrinsically flat because it can be cut and unrolled onto a plane without stretching.
- A saddle is curved.
- A cone is flat away from its tip; its curvature is concentrated at the singular tip.

Curvature is the obstruction to flattening a metric by a change of coordinates. Tidal acceleration is the obstruction to eliminating gravity by a change to a freely falling frame. These are not merely analogous statements. General relativity will identify spacetime curvature with the tidal content of the gravitational field.

## 1.11 What We Have Established

The argument has moved through three layers. Newtonian gravity first became a field theory:

$$\vec{A}(\vec{x}) = G \sum_i m_i \frac{\vec{R}_i}{R_i^3}, \quad \nabla \cdot \vec{A} = -4\pi G\rho. \quad (1.30)$$

Gauss's theorem then turned that local source equation into the exterior inverse-square field, the shell theorem, and the linear interior field of a uniform Earth.

Next, an accelerated coordinate system produced an apparent uniform field,  $\ddot{x}' = -g$ . Applying that equivalence to all local physics showed that light falls and supplied the correct scale, though not yet the full factor, for solar light deflection.

Finally, the limitation of the elevator became the opening to geometry. Uniform gravity can be removed locally; tidal forces cannot. Curved coordinates can describe a flat surface; intrinsic curvature cannot be removed. The next task is to make that connection between tidal gravity and spacetime curvature precise.